# Kavya Dave

Hamburg, Germany  davekavya188@gmail.com  +49 176 86663592

linkedin.com/in/kavyadave


## Professional Summary

Master's student in Information and Communication Systems at Technische Universität Hamburg (TUHH) with academic experience in machine learning, image processing, and data-driven systems. Familiar with Python-based data analysis and foundational ML techniques using scikit-learn and PyTorch. Experienced in developing automation scripts and data workflows in Linux and containerized environments (Docker, CI/CD), with an interest in multimodal machine learning combining text, images, and structured data for engineering applications.

## Education

**Technische Universität Hamburg (TUHH)**, Hamburg — Oct 2022 – Ongoing

- Master of Science in Information and Communication Systems
- Relevant coursework: Image Processing
- Currently in VII Semester

**Gujarat Technological University**, Bhavnagar, India — Jun 2013 – May 2017

- Bachelor of Engineering in Electronics and Communication Engineering

## Experience

**Hiwi (Scientific Assistant)**, Fraunhofer CML, ASR Department – Hamburg — Aug 2024 – Ongoing

- Engineered and implemented comprehensive testing frameworks for both backend (Python/FastAPI) and frontend (JavaScript/React.js) systems within a multi-disciplinary team.
- Significantly enhanced system reliability by designing and executing structured test plans that achieved extensive coverage for core backend and frontend functionalities.
- Authored clear, step-by-step technical documentation for local environment setup and testing procedures, significantly reducing onboarding time for new developers.
- Developed robust troubleshooting guides for common issues (e.g., dependency errors, test failures), improving team efficiency and knowledge sharing.
- Currently researching and developing a methodology for interdependent service testing within Docker Compose to enhance the reliability and integration of containerized maritime applications.

**Hiwi**, Technische Universität Hamburg,Institute of Environmental Technology and Energy Economics – Hamburg — April 2025 – Dec 2025

- Built Python automation scripts to generate legal documents and fill interactive PDF forms using CSV data.
- Implemented placeholder replacement in Word (.DOCX) templates and automated PDF form filling with text fields & checkboxes.
- Documented setup, usage, and best practices, ensuring reproducibility and ease of use for non-technical users.
- Built a multi-scenario energy system analysis dashboard using Python, Dash, and Plotly, enabling comparison of system cost and capacity impacts of photocatalysis market penetration.
- Modeled electricity & hydrogen production, heating & power systems, hydrogen infrastructure, and synthetic fuels using structured JSON-based data pipelines.
- Implemented dual-axis comparative bar charts with synchronized zero alignment to accurately represent energy/capacity vs. annualized cost trade-offs.
- Designed modular plotting architecture supporting multiple subsystems with distinct units (TWh, GW, TWkm, $MtCO_2$,Bln. € $a^{-1}$).
- Ensured data consistency and scalability through strict key-mapping between scenario definitions and

visualization layers.

**Working Student (Research Assistant)**, Fraunhofer IAPT, LPBF Department – Hamburg — Jun 2023 – Feb 2024

- Conducted process analysis and technical reporting for additive manufacturing optimization projects
- Assisted with data visualization tasks and supported internal workshops and training programs

**Integration Engineer**, Synaptics Pvt Ltd – Bangalore, India — Oct 2018 – Sep 2020

- Automated system performance reports using Linux shell scripts, reducing manual effort and improving accuracy
- Developed Bash scripts to collect and summarize system analysis data for reporting
- Created and maintained technical documentation for development workflows

## Academic Projects

**Localization of mobile robots coupling building information modeling and WiFi-based positioning** — Aug 2025 – March 2026

- System Optimization: Used a Genetic Algorithm (MOGA) to optimize 3D Access Point placement for maximum coverage and best geometry (GDOP).
- Digital Twin Simulation: Developed a simulation in MATLAB using BIM data and Ray Tracing to model signal bounces off concrete, metal, and glass.
- Deep Learning: Developed and evaluated a deep learning model based on ResNet-18 architecture for robot localization using wireless signal features.
- Performance Analysis: Validated accuracy across different SNR levels and materials, identifying concrete as the most reliable environment for localization.

**Robot-based Inspection using Fiducial Markers** – Hamburg, Germany — Apr 2024 – Jul 2024

- Studied robot positioning and orientation using AprilTag-based fiducial markers within a ROS-based system.
- Contributed to system evaluation and documentation of inspection workflows
- Collaborated in a team environment to analyze potential industrial inspection applications

**Software testing** – TUHH , Germany — Apr 2024 – Jul 2024

- Evaluated and compared software tools and frameworks for performance, reliability, and scalability.
- Designed and deployed Python-based automation tools and agents for validation and testing workflows.
- Integrated automated checks into development pipelines to improve software quality and verification.

## Technical Skills

**Programming:** Python, MATLAB, Linux (Bash), JavaScript

**ML & Data:** Basic knowledge of scikit-learn, pandas, numpy, feature engineering, model evaluation (precision/recall/F1)

**Basic understanding of PyTorch and neural network architectures (e.g., CNNs)**

**Tools:** Git, Docker, CI/CD

**Soft Skills:** Strong communication, attention to detail, proactive cross-functional collaboration

**Languages:** English (fluent), German (basic), Gujarati (native), Hindi (fluent)

## Interests

Music, Scuba Diving, Traveling, Photography, Cricket